OOP - Assignment 3 - Project

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1 Design & Implementation

1.1 Are some fields or methods equivalent, even if they have slightly different names in your implementation of Enemy and Player?

Yes :

- x, y, image, width, height
- move(), shoot(), isHit() methods

1.2 Which one of the fields/ methods could be implemented in the Sprite class?

- x, y, image, width, height
- Get methods
- paint(Painting painting)

1.3 Do some methods have to be present in the Sprite class but cannot be implemented there?

- move()
- isHit()

2 Additional Lists

2.1 Based on your hierarchy, is it possible to store Enemy or Shot objects in the SpriteList? Why?

Yes, it is possible to store Enemy and Shot objects in a SpriteList because both Enemy and Shot are subclasses of Sprite. According to inheritance rules, an object of a subclass can be assigned to a variable of its superclass type.

2.2 What would be the problem if Enemy and Shot objects were stored in the SpriteList?

The main problem is that objects stored in a SpriteList are treated as Sprite objects. As a result, only the fields and methods defined in the Sprite class are directly accessible. Methods specific to Enemy or Shot can't be used without explicit type casting.